

Classification or Regression?

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Features, Labels, Targets

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace both checks for Classification and regression. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Classification and regression. Proposed explanation: “Numbers always mean regression”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Compare new evidence independently

For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes. Write actual and predicted category lists, count correct categories, calculate each absolute error and the mean error in minutes. State what each result measures.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

System A predicts class codes 0=oak, 1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

6. Solve the given case

Task A predicts healthy/spotted; B predicts next week's height in cm.

Choose task and output types.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Task A predicts whether a plant is fern or grass. Task B predicts its height in centimetres. Which is classification and which is regression?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 3 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Confirm whether Plant Classifier predicts a health category or a numerical severity score; document why.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

11. Transfer practice from the original programme

System A predicts class codes 0=oak, 1=maple. System B predicts leaf area in cm². Name the task types and explain why code 1 is not 'one more species' than code 0.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Classify the target meaning

At dispatch, identify the task for predicting route type coded 1 or 2, duration in whole minutes, and whether duration exceeds 10 minutes. Explain your choices.

My reasoning and evidence

13. W2 Trace both checks

For actual 14 minutes and estimate 12 minutes, calculate absolute error and both slow categories using greater than 10. Explain why a correct category does not mean an exact duration.

My reasoning and evidence

14. W3 Repair a category average

Indoor is coded 1 and outdoor is coded 2. Someone averages predictions and reports route type 1.5 as an accurate halfway category. Explain the problem and name an appropriate check.

My reasoning and evidence

15. W4 Compare new evidence independently

For three fictional print jobs, actual completion times are 4, 5 and 9 minutes; estimates are 6, 5 and 8 minutes in matching order. Slow means more than 5 minutes. Write actual and predicted category lists, count correct categories, calculate each absolute error and the mean error in minutes. State what each result measures.

My reasoning and evidence

16. W5 Specify a useful output

A user needs the arrival duration in minutes, but a report gives only slow or not slow. What task should be specified, and what information was lost by the category output?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Classification or Regression?

Distinguish predicting growth height from predicting a growth category.

Actual heights are [4, 7, 9] cm; estimates are [5, 6, 10] cm. Tall means strictly greater than 6 cm.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/3

Actual and estimated heights (paired order)

cm

actual1



estimate1



actual2



estimate2



actual3



estimate3



Bar length shows the supplied value; the marked baseline is zero.

Classification or Regression?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regression errors

Absolute errors are 1, 1, 1 cm; mean absolute error is 1 cm.



2. Actual categories

Actual classes are short, tall, tall.



3. Predicted categories

Estimated classes are short, short, tall: 2/3 correct.

CHANGE ONE THING

Change the category boundary to strictly greater than 7 cm.

Explain the changed result using the matching lesson.

Did regression MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Distinguish predicting growth height from predicting a growth category.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Estimated classes are short, short, tall: 2/3 correct.

1. Actual categories
2. Regression errors
3. Predicted categories

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Predicted categories Estimated classes are short, short, tall: 2/3 correct. Read the picture in order. Estimated classes are short, short, tall: 2/3 correct. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Change the category boundary to strictly greater than 7 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Actual and estimated categories both become short, short, tall: 3/3 correct.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Actual and estimated categories both become short, short, tall: 3/3 correct. The numerical estimates did not improve; the decision definition changed.

Repair a missing building block

Review Features, Labels, Targets (ai-4/2). Goal: Separate prediction inputs from observed targets. Short practice: Why can a historical table contain a field that is an invalid input? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
